Appendices to "Hamilton decompositions of the directed 3-torus $D(m)$: explicit constructions for all $m \geq 3$"

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A Route E return maps

Definition 4 appears in Section 4. This appendix records the closed-form return maps on P_0 derived from that definition in Appendix B and analyzed in Appendix C.

Return maps on P_0. Parameterize $P_0 = \{S = 0\}$ by

$$v(i,k) := (i, -i-k, k), \qquad (i,k) \in \mathbb{Z}_m^2.$$

For the explicit Route E direction assignment, define $R_c = f_c^m|_{P_0}$. (We write R_c rather than F_c to distinguish the associated m-step maps of Route E from the return maps F_c of the Kempe-swap coloring analyzed in Section 3.) The following piecewise formulas are derived directly from Definition 4 in Appendix B.

Case I: $m \equiv 0, 2 \pmod 6$**.**

$$R_0(i,k) = \begin{cases} (i-2,\ k) & (i,k) = (0,0), \\ (i-3,\ k+3) & (i,k) = (1, m-1), \\ (i-2,\ k+2) & (i,k) = (m-2, 0), \\ (i-1,\ k) & (i,k) = (m-1, 1), \\ (i-3,\ k+2) & i+k \equiv 1 \text{ and } (i,k) \neq (1,0), \\ (i-1,\ k+1) & ((k=0 \text{ and } 1 \leq i \leq m-3) \text{ or } (i,k) \in \{(0,m-1),(m-2,1)\}), \\ (i-2,\ k+1) & \text{otherwise.} \end{cases}$$

$$R_1(i,k) = \begin{cases} (i+2,\ k) & \big((i+k \equiv m-1 \text{ and } 1 \leq i \leq m-3) \\ & \quad \text{or } (i,k) \in \{(0,0),(m-2,0),(m-1,m-1)\}\big), \\ (i+1,\ k) & \big((i=0 \text{ and } 1 \leq k \leq m-2) \\ & \quad \text{or } (i,k) \in \{(1,m-1),(m-2,1)\}\big), \\ (i+1,\ k+1) & \text{otherwise.} \end{cases}$$

Case II: $m \equiv 4 \pmod 6$**.**

$$R_0(i,k) = \begin{cases} (i-2,\ k) & (i,k) = (0,0), \\ (i-1,\ k) & (i,k) = (m-1,1), \\ (i-3,\ k+3) & (i,k) \in \{(1,m-1),(2,m-2)\}, \\ (i-2,\ k+2) & \big((i=1 \text{ and } 0 \le k \le m-3) \text{ or } (i,k) \in \{(2,m-1),(m-2,0)\}\big), \\ (i-3,\ k+2) & i+k \equiv 1 \text{ and } (i,k) \notin \{(1,0),(2,m-1)\}, \\ (i-1,\ k+1) & \big((k=0 \text{ and } 2 \le i \le m-3) \\ & \quad \text{or } (i,k) \in \{(0,m-1),(1,m-2),(m-2,1)\}\big), \\ (i-2,\ k+1) & \text{otherwise.} \end{cases}$$

$$R_1(i,k) = \begin{cases} (i+2,\ k) & \big((i+k \equiv m-1 \text{ and } 2 \le i \le m-3) \\ & \quad \text{or } (i,k) \in \{(0,0),(1,0),(m-2,0),(m-1,m-1)\}\big), \\ (i+1,\ k) & \big((i=0 \text{ and } 1 \le k \le m-2) \text{ or } (i=1 \text{ and } 1 \le k \le m-1) \\ & \quad \text{or } (i,k) \in \{(2,m-2),(2,m-1),(m-2,1)\}\big), \\ (i+1,\ k+1) & \text{otherwise.} \end{cases}$$

Uniform for all even $m \ge 6$**.**

$$R_2(i,k) = \begin{cases} (i+1,\ k-3) & (i,k) = (2,0), \\ (i+2,\ k-3) & i+k \equiv 1 \text{ and } (i,k) \notin \{(1,0),(m-1,2)\}, \\ (i,\ k-1) & \big((k \equiv m-1-i \text{ and } i \not\equiv m-1) \text{ or } (i,k) = (m-1,m-1)\big), \\ (i,\ k-2) & \big((k=0 \text{ and } i \notin \{0,2,m-1\}) \text{ or } (i,k) = (m-1,1)\big), \\ (i+1,\ k-1) & \big((i=m-1 \text{ and } k \notin \{1,m-1\}) \text{ or } (i,k) = (0,0)\big), \\ (i+1,\ k-2) & \text{otherwise.} \end{cases}$$

B Deriving the Route E return maps from Definition 4

In this appendix we derive the return maps recorded in Appendix A directly from the low-layer Route E rule. The argument does not appeal to the defect-cycle structure; it tracks only the first three layers $S = 0,1,2$.

Proposition 1. *For every even $m \ge 6$, the return maps R_0, R_1, R_2 induced by Definition 4 are exactly the piecewise formulas displayed in Appendix A.*

Proof. Parameterize

$$P_0 = \{(i,j,k) \in \mathbb{Z}_m^3 : i+j+k \equiv 0\}$$

by

$$v(i,k) := (i, -i-k, k), \qquad (i,k) \in \mathbb{Z}_m^2.$$

Fix a color $c \in \{0,1,2\}$ and start from $v(i,k) \in P_0$. Write the first three color-c directions encountered on the layers $S = 0,1,2$ as

$$\omega_c(i,k) = (a_c, b_c, c_c) \in \{0,1,2\}^3.$$

After those three steps the orbit lies on layer $S = 3$. From there until the next return to P_0 all layers are canonical, so the remaining $m-3$ steps are all in direction c. Hence the whole return map is determined by the low-layer word $\omega_c(i,k)$.

If $N_r(\omega)$ denotes the number of occurrences of direction r in a word ω, then

$$R_0(i,k) = \big(i + N_0(\omega_0) - 3,\ k + N_2(\omega_0)\big),$$

$$R_1(i,k) = \big(i + N_0(\omega_1),\ k + N_2(\omega_1)\big),$$

$$R_2(i,k) = \big(i + N_0(\omega_2),\ k + N_2(\omega_2) - 3\big).$$

So it remains only to determine the low-layer words.

From Definition 4:

- on layer $S = 1$, the triple is $(1, 0, 2)$ if $i = 0$, and $(2, 0, 1)$ if $i \neq 0$;
- on layer $S = 2$, the triple is $(2, 1, 0)$ if $j = 0$, and $(0, 1, 2)$ if $j \neq 0$.

Therefore:

- for color 0, the layer-1 direction depends on whether the post-step-1 i-coordinate equals 0, and the layer-2 direction depends on whether the post-step-2 j-coordinate equals 0;
- for color 1, the directions on layers 1 and 2 are always 0 and 1;
- for color 2, the layer-1 direction depends on whether the post-step-1 i-coordinate equals 0, and the layer-2 direction depends on whether the post-step-2 j-coordinate equals 0.

Case I: $m \equiv 0, 2 \pmod 6$**.** In the (i,k)-coordinates on P_0, the layer-0 direction triples become

$$\begin{aligned}
A &:= \{(0,0)\} \cup \{(i,k) : i + k \equiv m - 1,\ 1 \le i \le m - 3\} \cup \{(m-1, m-1)\},\\
B &:= \{(0, m-1)\} \cup \{(i,0) : 1 \le i \le m - 3\} \cup \{(m-1, 1)\},\\
C &:= \{(0,k) : 1 \le k \le m - 2\} \cup \{(1, m-1)\},\\
D &:= \{(m-2, 1)\}, \qquad E := \{(m-2, 0)\},
\end{aligned}$$

corresponding respectively to the triples $102, 021, 210, 012, 201$, while the remaining points

$$F := \mathbb{Z}_m^2 \setminus (A \cup B \cup C \cup D \cup E)$$

carry the triple 120.

Color 1. On layers 1 and 2, color 1 always uses directions 0 and 1. So the low-layer word is always of the form $(a, 0, 1)$, where a is the layer-0 color-1 direction. Thus:

- if $a = 0$, then $\omega_1 = 001$ and $R_1(i,k) = (i + 2, k)$;
- if $a = 1$, then $\omega_1 = 101$ and $R_1(i,k) = (i + 1, k)$;
- if $a = 2$, then $\omega_1 = 201$ and $R_1(i,k) = (i + 1, k + 1)$.

Reading off the sets on which the layer-0 color-1 direction equals 0, 1, or 2 gives exactly the Case I formula for R_1 in Appendix A.

Color 0. For color 0, we determine the word $\omega_0 = (a_0, b_0, c_0)$ algebraically. Step 1 goes in direction a_0, producing $i_1 = i + \mathbf{1}_{a_0=0}$. The layer-1 rule gives $b_0 = 1$ if $i_1 \equiv 0 \pmod{m}$, and 2 otherwise. Step 2 goes in direction b_0, producing

$$j_2 = j + \mathbf{1}_{a_0=1} + \mathbf{1}_{b_0=1}, \qquad j \equiv -i - k \pmod{m}.$$

The layer-2 rule gives $c_0 = 2$ if $j_2 \equiv 0 \pmod{m}$, and 0 otherwise. Table 1 executes this arithmetic trace on each region of P_0. Grouping the start points by the final displacement yields exactly the displayed Case I formula for R_0.

Table 1: Derivation of low-layer words ω_0 for color 0, Case I ($m \equiv 0, 2 \pmod{6}$).

Set	Condition / point	a_0	$i_1 = i + \mathbf{1}_{a_0=0}$	b_0	$j_2 = j + \mathbf{1}_{a_0=1} + \mathbf{1}_{b_0=1}$	c_0	ω_0	Disp. $R_0(i,k)$
A	$(0,0)$	1	$0 \Rightarrow$	1	$0+1+1=2 \not\equiv 0 \Rightarrow$	0	110	$(i-2,k)$
	$A \setminus \{(0,0)\}$	1	$i \not\equiv 0 \Rightarrow$	2	$j+1+0 \not\equiv 0 \Rightarrow$	0	120	$(i-2,k+1)$
B	$(m-1,1)$	0	$0 \Rightarrow$	1	$0+0+1=1 \not\equiv 0 \Rightarrow$	0	010	$(i-1,k)$
	$B \setminus \{(m-1,1)\}$	0	$i+1 \not\equiv 0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	0	020	$(i-1,k+1)$
C	$(0,1)$	2	$0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	2	212	$(i-3,k+2)$
	$(1,m-1)$	2	$1 \not\equiv 0 \Rightarrow$	2	$0+0+0=0 \Rightarrow$	2	222	$(i-3,k+3)$
	remaining C	2	$0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	0	210	$(i-2,k+1)$
D	$(m-2,1)$	0	$m-1 \not\equiv 0 \Rightarrow$	2	$1+0+0=1 \not\equiv 0 \Rightarrow$	0	020	$(i-1,k+1)$
E	$(m-2,0)$	2	$m-2 \not\equiv 0 \Rightarrow$	2	$2+0+0=2 \not\equiv 0 \Rightarrow$	0	220	$(i-2,k+2)$
F	$i+k \equiv 1$ (equiv. $j \equiv -1$)	1	$i \not\equiv 0 \Rightarrow$	2	$-1+1+0=0 \Rightarrow$	2	122	$(i-3,k+2)$
	otherwise	1	$i \not\equiv 0 \Rightarrow$	2	$j+1+0 \not\equiv 0 \Rightarrow$	0	120	$(i-2,k+1)$

Color 2. Similarly, the layer-1 rule for color 2 gives $b_2 = 2$ if $i_1 \equiv 0 \pmod{m}$, and 1 otherwise. The layer-2 rule gives $c_2 = 0$ if $j_2 \equiv 0 \pmod{m}$, and 2 otherwise. Table 2 tracks this arithmetic. Grouping by displacement gives exactly the Case I formula for R_2.

Table 2: Derivation of low-layer words ω_2 for color 2, Case I ($m \equiv 0, 2 \pmod{6}$).

Set	Condition / point	a_2	$i_1 = i + \mathbf{1}_{a_2=0}$	b_2	$j_2 = j + \mathbf{1}_{a_2=1} + \mathbf{1}_{b_2=1}$	c_2	ω_2	Disp. $R_2(i,k)$
A	$(0,0)$	2	$0 \Rightarrow$	2	$0+0+0=0 \Rightarrow$	0	220	$(i+1,k-1)$
	$A \setminus \{(0,0)\}$	2	$i \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	212	$(i,k-1)$
B	$(0,m-1)$	1	$0 \Rightarrow$	2	$1+1+0=2 \not\equiv 0 \Rightarrow$	2	122	$(i,k-1)$
	$(2,0)$	1	$2 \not\equiv 0 \Rightarrow$	1	$-2+1+1=0 \Rightarrow$	0	110	$(i+1,k-3)$
	remaining B	1	$i \not\equiv 0 \Rightarrow$	1	$j+1+1 \not\equiv 0 \Rightarrow$	2	112	$(i,k-2)$
C	$(0,1)$	0	$1 \not\equiv 0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	0	010	$(i+2,k-3)$
	remaining C	0	$i+1 \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	012	$(i+1,k-2)$
D	$(m-2,1)$	2	$m-2 \not\equiv 0 \Rightarrow$	1	$1+0+1=2 \not\equiv 0 \Rightarrow$	2	212	$(i,k-1)$
E	$(m-2,0)$	1	$m-2 \not\equiv 0 \Rightarrow$	1	$2+1+1=4 \not\equiv 0 \Rightarrow$	2	112	$(i,k-2)$
F	$i=m-1$	0	$0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	2	022	$(i+1,k-1)$
	$i+k \equiv 1$ (equiv. $j \equiv -1$)	0	$i+1 \not\equiv 0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	0	010	$(i+2,k-3)$
	otherwise	0	$i+1 \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	012	$(i+1,k-2)$

Case II: $m \equiv 4 \pmod 6$**.** In the (i,k)-coordinates on P_0, the layer-0 direction triples become

$$\begin{aligned}
A &:= \{(0,0)\} \cup \{(i,k) : i+k \equiv m-1,\ 2 \le i \le m-3\} \cup \{(m-1, m-1)\},\\
B &:= \{(0,m-1)\} \cup \{(i,0) : 2 \le i \le m-3\} \cup \{(m-1,1)\},\\
C &:= \{(0,k) : 1 \le k \le m-2\} \cup \{(1,k) : 1 \le k \le m-3\} \cup \{(1,m-1)\} \cup \{(2,m-2),(2,m-1)\},\\
D &:= \{(1,m-2),(m-2,1)\}, \qquad E := \{(1,0),(m-2,0)\},
\end{aligned}$$

corresponding again to the triples $102, 021, 210, 012, 201$, with 120 on the remaining points

$$F := \mathbb{Z}_m^2 \setminus (A \cup B \cup C \cup D \cup E).$$

Color 1**.** Exactly the same low-layer argument applies. In Case II the layer-0 color-1 direction equals 0 on

$$(i+k \equiv m-1,\ 2 \le i \le m-3) \text{ or } (i,k) \in \{(0,0),(1,0),(m-2,0),(m-1,m-1)\},$$

it equals 1 on

$$(i=0,\ 1 \le k \le m-2) \text{ or } (i=1,\ 1 \le k \le m-1) \text{ or } (i,k) \in \{(2,m-2),(2,m-1),(m-2,1)\},$$

and it equals 2 otherwise. This is exactly the Case II formula for R_1.

Color 0**.** Applying the same coordinate checks over the Case II partition generates Table 3. Using the identical word-to-displacement dictionary from Case I verifies the displayed Case II formula for R_0.

Table 3: Derivation of low-layer words ω_0 for color 0, Case II ($m \equiv 4 \pmod 6$).

Set	Condition / point	a_0	$i_1 = i + \mathbf{1}_{a_0=0}$	b_0	$j_2 = j + \mathbf{1}_{a_0=1} + \mathbf{1}_{b_0=1}$	c_0	ω_0	Disp. $R_0(i,k)$
A	$(0,0)$	1	$0 \Rightarrow$	1	$0+1+1=2 \not\equiv 0 \Rightarrow$	0	110	$(i-2,k)$
	$A \setminus \{(0,0)\}$	1	$i \not\equiv 0 \Rightarrow$	2	$j+1+0 \not\equiv 0 \Rightarrow$	0	120	$(i-2,k+1)$
B	$(m-1,1)$	0	$0 \Rightarrow$	1	$0+0+1=1 \not\equiv 0 \Rightarrow$	0	010	$(i-1,k)$
	$B \setminus \{(m-1,1)\}$	0	$i+1 \not\equiv 0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	0	020	$(i-1,k+1)$
C	$(0,1)$	2	$0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	2	212	$(i-3,k+2)$
	$(1,m-1)$, $(2,m-2)$	2	$i \not\equiv 0 \Rightarrow$	2	$0+0+0=0 \Rightarrow$	2	222	$(i-3,k+3)$
	$(1,k)$ with $1 \le k \le m-3$, and $(2,m-1)$	2	$i \not\equiv 0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	0	220	$(i-2,k+2)$
	$(0,k)$ with $2 \le k \le m-2$	2	$0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	0	210	$(i-2,k+1)$
D	$(1,m-2)$, $(m-2,1)$	0	$i+1 \not\equiv 0 \Rightarrow$	2	$1+0+0=1 \not\equiv 0 \Rightarrow$	0	020	$(i-1,k+1)$
E	$(1,0)$, $(m-2,0)$	2	$i \not\equiv 0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	0	220	$(i-2,k+2)$
F	$i+k \equiv 1$ (equiv. $j \equiv -1$)	1	$i \not\equiv 0 \Rightarrow$	2	$-1+1+0=0 \Rightarrow$	2	122	$(i-3,k+2)$
	otherwise	1	$i \not\equiv 0 \Rightarrow$	2	$j+1+0 \not\equiv 0 \Rightarrow$	0	120	$(i-2,k+1)$

Color 2**.** The analogous evaluation for color 2 is tracked in Table 4. Grouping the outputs matches the uniform formula for R_2 displayed in Appendix A.

Table 4: Derivation of low-layer words ω_2 for color 2, Case II ($m \equiv 4 \pmod 6$).

Set	Condition / point	a_2	$i_1 = i + \mathbf{1}_{a_2=0}$	b_2	$j_2 = j + \mathbf{1}_{a_2=1} + \mathbf{1}_{b_2=1}$	c_2	ω_2	Disp. $R_2(i,k)$
A	$(0,0)$	2	$0 \Rightarrow$	2	$0+0+0=0 \Rightarrow$	0	220	$(i+1,k-1)$
	$A \setminus \{(0,0)\}$	2	$i \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	212	$(i,k-1)$
B	$(0,m-1)$	1	$0 \Rightarrow$	2	$1+1+0=2 \not\equiv 0 \Rightarrow$	2	122	$(i,k-1)$
	$(2,0)$	1	$2 \not\equiv 0 \Rightarrow$	1	$-2+1+1=0 \Rightarrow$	0	110	$(i+1,k-3)$
	remaining B	1	$i \not\equiv 0 \Rightarrow$	1	$j+1+1 \not\equiv 0 \Rightarrow$	2	112	$(i,k-2)$
C	$(0,1)$, $(2,m-1)$	0	$i+1 \not\equiv 0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	0	010	$(i+2,k-3)$
	remaining C	0	$i+1 \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	012	$(i+1,k-2)$
D	$(1,m-2)$, $(m-2,1)$	2	$i \not\equiv 0 \Rightarrow$	1	$1+0+1=2 \not\equiv 0 \Rightarrow$	2	212	$(i,k-1)$
E	$(1,0)$, $(m-2,0)$	1	$i \not\equiv 0 \Rightarrow$	1	$j+1+1 \not\equiv 0 \Rightarrow$	2	112	$(i,k-2)$
F	$i = m-1$	0	$0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	2	022	$(i+1,k-1)$
	$i+k \equiv 1$ (equiv. $j \equiv -1$)	0	$i+1 \not\equiv 0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	0	010	$(i+2,k-3)$
	otherwise	0	$i+1 \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	012	$(i+1,k-2)$

This proves all return-map formulas. □

C Hamiltonicity of the Route E return maps

C.1 Setup

The first-return framework of Section 4 applies uniformly to all three colors. The main text proves color 2 in full; this appendix records the remaining itinerary computations for colors 1 and 0. For color 1, Propositions 2 and 4 derive the first-return data; Corollary 3 establishes Hamiltonicity. For color 0, Propositions 6 and 8 do the same, with Corollary 6 completing the proof. The working coordinates and defect geometries for each color are summarized in Table 2.

The next lemma packages the cycle step that remains after the closed-form formulas for the first-return maps have been derived.

Lemma 1 (successor on cyclic blocks)**.** *Let X be a finite set written cyclically as*

$$X = (B_1 \mid B_2 \mid \cdots \mid B_r),$$

where each block $B_j = (x_{j,0}, x_{j,1}, \ldots, x_{j,\ell_j})$ is linearly ordered. Assume $T : X \to X$ satisfies

$$T(x_{j,s}) = x_{j,s+1} \quad (0 \le s < \ell_j), \qquad T(x_{j,\ell_j}) = x_{j+1,0},$$

with the block index j taken cyclically modulo r. Then T is a single $|X|$-cycle.

Proof. The hypotheses say exactly that T moves forward by one step in the cyclic concatenation

$$(x_{1,0}, x_{1,1}, \ldots, x_{1,\ell_1}, x_{2,0}, \ldots, x_{r,\ell_r}).$$

Hence every point lies on the same orbit, and that orbit has length $|X|$. □

In the propositions below we use Lemma 1 to repackage the induced lane maps T_c as successor permutations on cyclic concatenations of a bounded number of long arithmetic runs. This does

not replace the derivation of the closed-form formulas for T_c and ρ_c or the return-time sums; it only explains why the cycle structure is so simple once those formulas are known. When one of the displayed arithmetic runs is empty for the smallest admissible values of m, it is omitted from the cyclic concatenation.

C.2 Color 2

The color-2 return map is proved in full in the main text (Section 4.2). Proposition 4 derives the first-return data, and Corollary 3 establishes that R_2 is a single m^2-cycle for all even $m \geq 6$. We do not repeat those orbit traces here.

C.3 Color 1

By Lemma 8, the bulk coordinates for color 1 are $\Phi_1(i,k) = (u,t) = (i-k,\, k)$, in which the generic branch is $(u,t) \mapsto (u, t+1)$. By Lemma 9(i), the defect set lies on the affine lines $u+t=0$ and $u+2t = m-1$ (and, in Case II, also $u+t=1$), together with finitely many isolated points. Following the same pattern, it suffices to list the defect components met before first return and sum the lane increments.

We use a first-return analysis on the transversal

$$L_1 = \{(x,0) : x \in \mathbb{Z}_m\} \subseteq P_0.$$

Let $\rho_1(x)$ be the first return time of R_1 from $(x,0)$ to L_1, and define

$$R_1^{\rho_1(x)}(x,0) = (T_1(x), 0).$$

In the original (i,k)-coordinates, $u = i-k$ is the lane coordinate. A bulk move $(i,k) \mapsto (i+1, k+1)$ preserves u. The two defect branches act as stalls: $(i,k) \mapsto (i+1,k)$ increases u by 1 (defect line $u+t=0$, i.e. $i=0$), and $(i,k) \mapsto (i+2,k)$ increases u by 2 (defect line $u+2t = m-1$, i.e. diagonal $i+k = m-1$). Starting from $(x,0)$, the net change in i at first return is exactly the total lane increment contributed by the stalls.

C.3.1 Case I: $m \equiv 0, 2 \pmod 6$

Proposition 2. *Assume $m \equiv 0, 2 \pmod 6$. Then the first-return map and return times on L_1 are*

$$T_1(x) = \begin{cases} 2, & x = 0, \\ x+3, & 1 \leq x \leq m-4, \\ 1, & x = m-3, \\ 0, & x = m-2, \\ 3, & x = m-1, \end{cases} \qquad \rho_1(x) = \begin{cases} 1, & x = 0 \text{ or } x = m-2, \\ m+2, & 1 \leq x \leq m-4, \\ m+3, & x = m-3 \text{ or } x = m-1. \end{cases}$$

Proof. The closed-form map R_1 has three types of moves:

- bulk $(i,k) \mapsto (i+1, k+1)$,
- $+1$ stalls on $i=0$ together with $(1, m-1)$ and $(m-2, 1)$,
- $+2$ stalls on the diagonal $i+k = m-1$ together with $(0,0)$, $(m-2,0)$ and $(m-1, m-1)$.

The immediate cases are read directly from the formula:

$$(0,0) \mapsto (2,0), \qquad (m-2,0) \mapsto (0,0).$$

Hence

$$T_1(0) = 2, \quad \rho_1(0) = 1, \qquad T_1(m-2) = 0, \quad \rho_1(m-2) = 1.$$

Now fix $1 \le x \le m-4$.

Odd x. On the lane $u = x$ one reaches the diagonal $i + k = m - 1$ once before k wraps, namely at

$$k_1 = \frac{m-1-x}{2}, \qquad (i,k) = \left(\frac{m-1+x}{2}, \frac{m-1-x}{2}\right).$$

That point is in the $+2$ branch, so the lane becomes $x+2$. On the new lane $u = x+2$, the next defect before wrap is the vertical point $(0, m-x-2)$, which is a $+1$ stall, so the lane then becomes $x+3$. On that lane there is no diagonal defect before wrap, because the system

$$i - k = x + 3, \qquad i + k = m - 1$$

would force

$$k = \frac{m-x-4}{2},$$

which is not an integer when x is odd. Its vertical defect is at height $m-x-3$, already behind the current value $k = m-x-2$. Thus there are exactly two stalls, with total lane increment $2+1=3$. Therefore

$$T_1(x) = x+3, \qquad \rho_1(x) = m+2.$$

Even x. The lane $u = x$ has no diagonal hit. Its first defect is the vertical point $(0, m-x)$, which contributes a $+1$ stall and moves the orbit to lane $x+1$. That new lane is odd, so before wrap it hits the diagonal once, at

$$k_2 = m - \frac{x}{2} - 1,$$

and that diagonal hit is a $+2$ stall. So the lane becomes $x+3$. On lane $x+3$ the vertical defect is at height $m-x-3$, while the diagonal equation

$$i - k = x + 3, \qquad i + k = m - 1$$

gives the candidate height

$$k = \frac{m-x-4}{2}.$$

Both heights are strictly below the current value $k_2 = m - \frac{x}{2} - 1$. Hence there are no more stalls before return. Again the total lane increment is $1+2=3$, and the number of stalls is 2, giving

$$T_1(x) = x+3, \qquad \rho_1(x) = m+2.$$

It remains to treat the two boundary values.

The point $x = m-3$. The orbit begins

$$(m-3,0) \to (m-2,1) \to (m-1,1) \to (0,2) \to (1,2).$$

Thus it first gets a $+1$ stall at $(m-2,1)$ and then another $+1$ stall at $(0,2)$. At that point the orbit lies on the lane $u = i - k \equiv m-1$. The unique diagonal defect on this lane before k wraps is

$$\left(\frac{m}{2} - 1, \frac{m}{2}\right),$$

so the orbit continues

$$(1,2) \xrightarrow{\text{bulk } \frac{m}{2}-2} \left(\frac{m}{2}-1, \frac{m}{2}\right) \xrightarrow{+2} \left(\frac{m}{2}+1, \frac{m}{2}\right) \xrightarrow{\text{bulk } \frac{m}{2}} (1,0).$$

Hence the third stall is this $+2$ diagonal stall, there are exactly three stalls in total, and therefore

$$T_1(m-3) = 1, \qquad \rho_1(m-3) = m+3.$$

The point $x = m-1$. The orbit begins

$$(m-1,0) \to (0,1) \to (1,1),$$

so it gets a $+1$ stall at $(0,1)$. Later it reaches $(m-1, m-1)$, giving a $+2$ stall, and then $(1, m-1)$, giving the final $+1$ stall. At that point the orbit is on lane 3 with current clock value $k = m-1$. The vertical defect on lane 3 is at height $m-3$, while a diagonal defect on lane 3 would require

$$k = \frac{m-4}{2},$$

so both possible defect heights lie strictly behind the current value $m-1$. Hence there are no more defects before return. Therefore the total lane increment is 4 and the number of stalls is 3, so

$$T_1(m-1) = 3, \qquad \rho_1(m-1) = m+3.$$

This proves the stated formulas. □

Proposition 3 (splicing of arithmetic runs for color 1, Case I)**.** *Assume $m \equiv 0, 2 \pmod 6$. Then T_1 is the successor permutation on a cyclic order obtained from three long arithmetic runs together with $O(1)$ bridge vertices. More precisely,*

$$(0, 2, 5, \ldots, m-3 \mid 1, 4, 7, \ldots, m-1 \mid 3, 6, 9, \ldots, m-2)$$

when $m \equiv 2 \pmod 6$, and

$$(0, 2, 5, \ldots, m-1 \mid 3, 6, 9, \ldots, m-3, 1 \mid 4, 7, 10, \ldots, m-2)$$

when $m \equiv 0 \pmod 6$. Consequently T_1 is a single m-cycle.

Proof. By Proposition 2,

$$T_1(0) = 2, \qquad T_1(x) = x+3 \quad (1 \le x \le m-4), \qquad T_1(m-3) = 1, \qquad T_1(m-2) = 0, \qquad T_1(m-1) = 3.$$

Hence the generic action is translation by $+3$, and only the endpoint values $m-3, m-2, m-1$ alter the itinerary. If $m \equiv 2 \pmod 6$, the three $+3$-runs are

$$2, 5, \ldots, m-3, \qquad 1, 4, 7, \ldots, m-1, \qquad 3, 6, 9, \ldots, m-2,$$

with the special values splicing them in the order

$$0 \to 2, \qquad m-3 \to 1, \qquad m-1 \to 3, \qquad m-2 \to 0.$$

If $m \equiv 0 \pmod 6$, the same generic rule produces the runs

$$2, 5, \ldots, m-1, \qquad 3, 6, 9, \ldots, m-3, \qquad 1, 4, 7, \ldots, m-2,$$

and the endpoint values splice them in the order

$$0 \to 2, \qquad m-1 \to 3, \qquad m-3 \to 1, \qquad m-2 \to 0.$$

In either congruence class this is exactly the displayed cyclic order, so Lemma 1 applies. □

Corollary 1. *If $m \equiv 0, 2 \pmod 6$, then R_1 is a single m^2-cycle.*

Proof. Proposition 3 shows that T_1 is a single m-cycle. Also,

$$\sum_{x \in \mathbb{Z}_m} \rho_1(x) = 2 + (m-4)(m+2) + 2(m+3) = m^2.$$

Now apply Lemma 7. □

C.3.2 Case II: $m \equiv 4 \pmod 6$

Proposition 4. *Assume $m \equiv 4 \pmod 6$. Then the first-return map and return times on L_1 are*

$$T_1(x) = \begin{cases} 2, & x = 0, \\ 3, & x = 1, \\ 5, & x = 2, \\ x+2, & x \text{ even}, \ x \notin \{0, 2, m-2\}, \\ 0, & x = m-2, \\ x+6, & x \text{ odd}, \ x \notin \{1, m-3, m-1\}, \\ 4, & x = m-3, \\ 7, & x = m-1, \end{cases}$$

and

$$\rho_1(x) = \begin{cases} 1, & x \in \{0, 1, m-2\}, \\ m+3, & x = 2, \\ m+2, & x \text{ even}, \ x \notin \{0, 2, m-2\}, \\ m+4, & x \text{ odd}, \ x \notin \{1, m-3, m-1\}, \\ m+6, & x \in \{m-3, m-1\}. \end{cases}$$

Proof. In Case II the closed-form map has

- bulk $(i, k) \mapsto (i+1, k+1)$,
- $+1$ stalls on the lines $i = 0$ and $i = 1$, together with the special points $(2, m-2)$, $(2, m-1)$, $(m-2, 1)$,
- $+2$ stalls on the diagonal $i + k = m - 1$ for $2 \le i \le m-3$, together with the special points $(0, 0)$, $(1, 0)$, $(m-2, 0)$, $(m-1, m-1)$.

The immediate cases are read directly from the formula:

$$(0,0) \mapsto (2,0), \qquad (1,0) \mapsto (3,0), \qquad (m-2,0) \mapsto (0,0),$$

so

$$T_1(0) = 2, \quad T_1(1) = 3, \quad T_1(m-2) = 0,$$

and all three return times equal 1.

For $x = 2$, the lane $u = 2$ has no diagonal defect. The orbit runs generically to $(0, m-2)$, then encounters three consecutive $+1$ stalls:

$$(2,0) \xrightarrow{\text{bulk}^{m-2}} (0, m-2) \xrightarrow{+1} (1, m-2) \xrightarrow{+1} (2, m-2) \xrightarrow{+1} (3, m-2).$$

From there two bulk moves return to L_1. Hence

$$T_1(2) = 5, \qquad \rho_1(2) = m + 3.$$

Now let x be even with $4 \le x \le m - 4$. The lane $u = x$ has no diagonal defect. Its first defect is $(0, m - x)$, followed immediately by the $i = 1$ defect at the same height:

$$(x, 0) \xrightarrow{\text{bulk}^{m-x}} (0, m - x) \xrightarrow{+1} (1, m - x) \xrightarrow{+1} (2, m - x).$$

Because $x \neq 2$, the point $(2, m - x)$ is not one of the special $+1$ points. The orbit is now on the even lane $x + 2$. On that lane there is no diagonal defect before wrap, and the remaining vertical defects occur at heights $m - x - 2$ and $m - x - 1$, both already behind the current value $k = m - x$. So no further defect occurs before the return to L_1, and therefore

$$T_1(x) = x + 2, \qquad \rho_1(x) = m + 2.$$

Next let x be odd with $3 \le x \le m - 7$. Define

$$d_x = \frac{m - 1 - x}{2}, \qquad e_x = m - x - 2, \qquad f_x = m - \frac{x + 5}{2}.$$

Because x is odd, the first diagonal defect on lane $u = x$ occurs at height d_x, namely at

$$D_1 = \left(\frac{m - 1 + x}{2}, \frac{m - 1 - x}{2} \right).$$

So the orbit first performs one $+2$ stall there. It then lies on lane $x + 2$, whose next defects are the vertical points

$$V_0 = (0, e_x), \qquad V_1 = (1, e_x),$$

which contribute two consecutive $+1$ stalls. The orbit then lies on the odd lane $x + 4$. On that lane the lower diagonal height is already behind the current clock value, while the upper diagonal height is exactly f_x. Thus the next defect is the diagonal point

$$D_2 = \left(\frac{x + 3}{2}, m - \frac{x + 5}{2} \right),$$

which contributes the second $+2$ stall. After this move the orbit lies on lane $x + 6$. On that lane the remaining vertical defect heights are $m - x - 6$ and $m - x - 5$, and the remaining diagonal height before wrap is $m - (x + 7)/2$; all of these lie behind the current value $k = f_x$. So no further defect occurs before the return to L_1. In total the orbit has two $+2$ stalls and two $+1$ stalls, hence total lane increment 6 and return time $m + 4$:

$$T_1(x) = x + 6, \qquad \rho_1(x) = m + 4.$$

For the endpoint $x = m - 5$, the same initial defect pattern occurs: one $+2$ stall on the first diagonal hit, then two consecutive $+1$ stalls on the lines $i = 0$ and $i = 1$, and then one more $+2$ stall on the second diagonal hit. After that second diagonal stall the orbit lies on lane 1 at height $m/2$. The lower diagonal point on lane 1 is already behind the current clock value, and the only remaining intersection with the diagonal before wrap is $(0, m - 1)$. This point is not a defect: it is excluded both from the $i = 0$ $+1$ branch and from the diagonal $+2$ branch. So the next step is the bulk move

$$(0, m - 1) \mapsto (1, 0),$$

which is the first return to L_1. Thus

$$T_1(m-5) = 1 = (m-5)+6, \qquad \rho_1(m-5) = m+4.$$

For $x = m-3$, the orbit is

$$\begin{aligned}(m-3,0) &\xrightarrow{\text{bulk}} (m-2,1) \xrightarrow{+1} (m-1,1) \xrightarrow{\text{bulk}} (0,2) \xrightarrow{+1} (1,2)\\ &\xrightarrow{+1} (2,2) \xrightarrow{\text{bulk}^{m-3}} (m-1,m-1) \xrightarrow{+2} (1,m-1)\\ &\xrightarrow{+1} (2,m-1) \xrightarrow{+1} (3,m-1) \xrightarrow{\text{bulk}} (4,0).\end{aligned}$$

So the lane increment is $1+1+1+2+1+1 = 7$ and the return time is $m+6$. Hence

$$T_1(m-3) = 4, \qquad \rho_1(m-3) = m+6.$$

For $x = m-1$, the orbit is

$$\begin{aligned}(m-1,0) &\xrightarrow{\text{bulk}} (0,1) \xrightarrow{+1} (1,1) \xrightarrow{+1} (2,1) \xrightarrow{\text{bulk}^{\frac{m}{2}-2}} \left(\frac{m}{2}, \frac{m}{2}-1\right)\\ &\xrightarrow{+2} \xrightarrow{\text{bulk}^{\frac{m}{2}-2}} (0,m-3) \xrightarrow{+1} (1,m-3) \xrightarrow{+1} (2,m-3)\\ &\xrightarrow{+2} (4,m-3) \xrightarrow{\text{bulk}^3} (7,0).\end{aligned}$$

Thus the orbit has four $+1$ stalls and two $+2$ stalls, for total lane increment 8 and return time $m+6$. Therefore

$$T_1(m-1) = 7, \qquad \rho_1(m-1) = m+6.$$

□

Proposition 5 (splicing of arithmetic runs for color 1, Case II)**.** *Assume $m \equiv 4 \pmod 6$. Then T_1 is the successor permutation on a cyclic order obtained from three long arithmetic runs together with $O(1)$ bridge vertices:*

$$(0,2,5,11,\ldots,m-5,1 \mid 3,9,15,\ldots,m-1,7,13,\ldots,m-3 \mid 4,6,8,\ldots,m-2).$$

Consequently T_1 is a single m-cycle.

Proof. Proposition 4 shows that, away from the boundary values

$$0,1,2,m-3,m-2,m-1,$$

the map acts by

$$x \mapsto x+2 \quad \text{on even residues}, \qquad x \mapsto x+6 \quad \text{on odd residues}.$$

The special values then splice these generic arithmetic runs as follows:

$$0 \to 2 \to 5,$$

after which the odd $+6$-run continues through

$$5,11,\ldots,m-5$$

and ends at

$$m-5 \to 1.$$

Next

$$1 \to 3,$$

after which the odd +6-rule produces the second odd run

$$3, 9, 15, \ldots, m-1, 7, 13, \ldots, m-3,$$

and the boundary value

$$m-3 \to 4$$

enters the even run

$$4, 6, 8, \ldots, m-2.$$

Finally

$$m-2 \to 0$$

closes the cycle. Thus T_1 is exactly the successor map on the displayed cyclic order, so Lemma 1 applies. □

Corollary 2. *If $m \equiv 4 \pmod 6$, then R_1 is a single m^2-cycle.*

Proof. Proposition 5 shows that T_1 is a single m-cycle. Also,

$$\sum_{x \in \mathbb{Z}_m} \rho_1(x) = 3 + (m+3) + \frac{m-6}{2}(m+2) + \frac{m-6}{2}(m+4) + 2(m+6) = m^2.$$

Therefore Lemma 7 applies. □

Corollary 3. *For every even $m \geq 6$, the return map R_1 is a single m^2-cycle on P_0.*

C.4 Color 0

By Lemma 8, the bulk coordinates for color 0 are $\Phi_0(i,k) = (u,t) = (i+2k,\, k)$, in which the generic branch is $(u,t) \mapsto (u, t+1)$. By Lemma 9(iii), the defect set lies on the lines $t = 0$ and $u = t+1$ (and, in Case II, also $u = 1+2t$), together with finitely many isolated points. Since the bulk frame already uses the convenient coordinates $(x,y) = (u,t) = (i+2k,\, k)$, we work directly in this frame.

The transversal is

$$L_0 = \{(x,0) : x \in \mathbb{Z}_m\}.$$

The generic move freezes x and increments y, so y is literally a clock while the exceptional branches change only the lane coordinate x.

C.4.1 Case I: $m \equiv 0, 2 \pmod 6$

Lemma 2. *Assume $m \equiv 0, 2 \pmod 6$. In the coordinates $(x, y) = (i + 2k, k)$, the return map R_0 becomes*

$$R_0(x,y) = \begin{cases} (m-2, 0), & (x,y) = (0,0), \\ (2,2), & (x,y) = (m-1, m-1), \\ (0,2), & (x,y) = (m-2, 0), \\ (0,1), & (x,y) = (1,1), \\ (x+1, y+2), & x = y+1,\ (x,y) \neq (1,0), \\ (x+1, y+1), & (y = 0,\ 1 \leq x \leq m-3) \text{ or } (x,y) \in \{(0,1), (m-2, m-1)\}, \\ (x, y+1), & \text{otherwise.} \end{cases}$$

All arithmetic is modulo m.

Proof. Substitute $i = x - 2y$ and $k = y$ into the Case I formula for R_0 in Appendix A and simplify branch by branch. The generic move $(i, k) \mapsto (i - 2, k + 1)$ becomes $(x, y) \mapsto (x, y + 1)$, and the six exceptional branches transform exactly as displayed. □

Write

$$G(x, y) = (x, y + 1)$$

for the generic branch in Lemma 2. The six exceptional moves will be denoted P, Q, R, S, A, B in the order displayed in the lemma: P and Q are isolated-point defects at $(0, 0)$ and $(m - 1, m - 1)$; R maps $(m - 2, 0)$ to $(0, 2)$; S maps $(1, 1)$ to $(0, 1)$; A is the defect on the line $u = t + 1$ (i.e. $x = y + 1$); B is the defect on the line $t = 0$ (i.e. $y = 0$) together with the boundary points $(0, 1)$ and $(m - 2, m - 1)$.

Proposition 6. *Assume $m \equiv 0, 2 \pmod 6$. Then the first-return map on L_0 is*

$$T_0(x) = \begin{cases} m-2, & x = 0, \\ x+2, & 1 \leq x \leq m-5, \\ m-1, & x = m-4, \\ 2, & x = m-3, \\ 1, & x = m-2, \\ 0, & x = m-1, \end{cases}$$

and the return times are

$$\rho_0(x) = \begin{cases} 1, & x = 0, \\ m-1, & 1 \leq x \leq m-4 \text{ or } x = m-1, \\ 2m-3, & x = m-3, \\ 2m-1, & x = m-2. \end{cases}$$

Proof. Fix $x \in \mathbb{Z}_m$ and start from $(x, 0) \in L_0$.

Case $x = 0$. By branch P,

$$(0,0) \xrightarrow{P} (m-2, 0),$$

so $T_0(0) = m - 2$ and $\rho_0(0) = 1$.

Case $1 \le x \le m-5$**.** The orbit is

$$(x,0) \xrightarrow{B} (x+1,1) \xrightarrow{G^{x-1}} (x+1,x) \xrightarrow{A} (x+2,x+2) \xrightarrow{G^{m-x-2}} (x+2,0).$$

Indeed, on the lane $x+1$ the first defect above $y=1$ is the A-point $(x+1,x)$. After that jump the orbit lies on lane $x+2$ at height $y=x+2$. On this lane the only possible A-point is $(x+2,x+1)$, whose height $x+1$ is already behind the current clock value. The special points P,Q,R,S lie on different lanes, and the B-branch occurs only at $y=0$ or at the isolated points $(0,1)$ and $(m-2,m-1)$, none of which lies ahead on lane $x+2$. Thus every remaining step before $y=0$ is generic. Therefore

$$T_0(x) = x+2, \qquad \rho_0(x) = 1+(x-1)+1+(m-x-2) = m-1.$$

Case $x = m-4$**.** Here

$$(m-4,0) \xrightarrow{B} (m-3,1) \xrightarrow{G^{m-5}} (m-3,m-4) \xrightarrow{A} (m-2,m-2) \xrightarrow{G} (m-2,m-1) \xrightarrow{B} (m-1,0).$$

So

$$T_0(m-4) = m-1, \qquad \rho_0(m-4) = 1+(m-5)+1+1+1 = m-1.$$

Case $x = m-3$**.** The orbit is

$$(m-3,0) \xrightarrow{B} (m-2,1) \xrightarrow{G^{m-4}} (m-2,m-3) \xrightarrow{A} (m-1,m-1) \xrightarrow{Q} (2,2) \xrightarrow{G^{m-2}} (2,0).$$

Hence

$$T_0(m-3) = 2, \qquad \rho_0(m-3) = 1+(m-4)+1+1+(m-2) = 2m-3.$$

Case $x = m-2$**.** The orbit is

$$(m-2,0) \xrightarrow{R} (0,2) \xrightarrow{G^{m-3}} (0,m-1) \xrightarrow{A} (1,1) \xrightarrow{S} (0,1) \xrightarrow{B} (1,2) \xrightarrow{G^{m-2}} (1,0).$$

Therefore

$$T_0(m-2) = 1, \qquad \rho_0(m-2) = 1+(m-3)+1+1+1+(m-2) = 2m-1.$$

Case $x = m-1$**.** The orbit is

$$(m-1,0) \xrightarrow{G^{m-2}} (m-1,m-2) \xrightarrow{A} (0,0).$$

So

$$T_0(m-1) = 0, \qquad \rho_0(m-1) = m-2+1 = m-1.$$

These cases exhaust $x \in \mathbb{Z}_m$. □

Proposition 7 (splicing of arithmetic runs for color 0, Case I)**.** *Assume* $m \equiv 0,2 \pmod 6$*. Then* T_0 *is the successor permutation on a cyclic order obtained from two long arithmetic runs together with* $O(1)$ *bridge vertices:*

$$(0, m-2 \mid 1,3,5,\ldots,m-3 \mid 2,4,6,\ldots,m-4 \mid m-1).$$

Consequently T_0 *is a single* m*-cycle.*

Proof. Proposition 6 gives

$$T_0(0) = m-2, \qquad T_0(x) = x+2 \quad (1 \le x \le m-5), \qquad T_0(m-4) = m-1,$$

$$T_0(m-3) = 2, \qquad T_0(m-2) = 1, \qquad T_0(m-1) = 0.$$

So the generic action is translation by $+2$ on the long odd and even runs

$$1, 3, 5, \ldots, m-3, \qquad 2, 4, 6, \ldots, m-4,$$

while the exceptional values splice them through the two-point block

$$0 \to m-2$$

and the one-point tail

$$m-1 \to 0.$$

More explicitly,

$$m-2 \to 1, \qquad m-3 \to 2, \qquad m-4 \to m-1, \qquad m-1 \to 0,$$

so T_0 is the successor map on the displayed cyclic order. Lemma 1 now applies. □

Corollary 4. *If $m \equiv 0, 2 \pmod 6$, then R_0 is a single m^2-cycle.*

Proof. Proposition 7 shows that T_0 is a single m-cycle. Also

$$\sum_{x \in \mathbb{Z}_m} \rho_0(x) = 1 + (m-3)(m-1) + (2m-3) + (2m-1) = m^2.$$

Lemma 7 now applies. □

C.4.2 Case II: $m \equiv 4 \pmod 6$

Lemma 3. *Assume $m \equiv 4 \pmod 6$. In the coordinates $(x, y) = (i + 2k, k)$, the return map R_0 becomes*

$$R_0(x,y) = \begin{cases} (m-2, 0), & (x,y) = (0,0), \\ (x+3, y+3), & (x,y) \in \{(m-1, m-1), (m-2, m-2)\}, \\ (x+2, y+2), & (x,y) = (m-2, 0) \text{ or } (x,y) = (0, m-1) \text{ or } x \equiv 1 + 2y \pmod m,\ 0 \le y \le m-3, \\ (0,1), & (x,y) = (1,1), \\ (x+1, y+2), & x = y+1,\ (x,y) \notin \{(1,0), (0, m-1)\}, \\ (x+1, y+1), & (y = 0,\ 2 \le x \le m-3) \text{ or } (x,y) \in \{(0,1), (m-3, m-2), (m-2, m-1)\}, \\ (x, y+1), & \text{otherwise.} \end{cases}$$

All arithmetic is modulo m.

Proof. Again substitute $i = x - 2y$ and $k = y$ into the Case II formula from Appendix A. The generic branch becomes $(x, y) \mapsto (x, y+1)$, and the exceptional branches simplify exactly as shown. □

As before write

$$G(x, y) = (x, y + 1)$$

for the generic move. For an odd lane $u \in \mathbb{Z}_m$ define the two R-heights

$$r_-(u) = \frac{u-1}{2}, \qquad r_+(u) = \frac{u-1}{2} + \frac{m}{2}.$$

These are exactly the two solutions $y \in \{0, \ldots, m-1\}$ of

$$u \equiv 1 + 2y \pmod{m}.$$

Proposition 8. *Assume $m \equiv 4 \pmod 6$. Then the first-return map on L_0 is given by*

$$T_0(0) = m-2, \qquad T_0(1) = 4, \qquad T_0(2) = 6, \qquad T_0(x) = x + 4 \ \ (3 \le x \le m-7),$$

together with

$$T_0(m-6) = 3, \quad T_0(m-5) = m-1, \quad T_0(m-4) = 0, \quad T_0(m-3) = 2, \quad T_0(m-2) = 5, \quad T_0(m-1) = 1,$$

and the return times are

$$\rho_0(0) = 1, \qquad \rho_0(x) = m-2 \quad (\textit{for } 1 \le x \le m-7 \textit{ or } x = m-4),$$

$$\rho_0(m-6) = 2m-4, \quad \rho_0(m-2) = 2m-4, \quad \rho_0(m-5) = m-1, \quad \rho_0(m-1) = m-1, \quad \rho_0(m-3) = 2m-3.$$

Proof. Fix $x \in \mathbb{Z}_m$ and start from $(x, 0) \in L_0$.

Case $x = 0$. By the special point branch,

$$(0, 0) \xrightarrow{P} (m-2, 0),$$

so $T_0(0) = m-2$ and $\rho_0(0) = 1$.

Case $x = 1$. The orbit is

$$(1, 0) \xrightarrow{R} (3, 2) \xrightarrow{A} (4, 4) \xrightarrow{G^{m-4}} (4, 0).$$

So

$$T_0(1) = 4, \qquad \rho_0(1) = 1 + 1 + (m-4) = m-2.$$

Case $x = 2$. The orbit is

$$(2, 0) \xrightarrow{B} (3, 1) \xrightarrow{R} (5, 3) \xrightarrow{G} (5, 4) \xrightarrow{A} (6, 6) \xrightarrow{G^{m-6}} (6, 0),$$

hence

$$T_0(2) = 6, \qquad \rho_0(2) = 1 + 1 + 1 + 1 + (m-6) = m-2.$$

Case x odd, $3 \le x \le m-7$. The first move is

$$(x, 0) \xrightarrow{B} (x+1, 1),$$

and the lane $x + 1$ is even, so there is no R-branch on that lane. The next defect is the A-point $(x + 1, x)$. Thus

$$(x, 0) \xrightarrow{B} (x+1, 1) \xrightarrow{G^{x-1}} (x+1, x) \xrightarrow{A} (x+2, x+2).$$

Now the orbit lies on the odd lane $x+2$. Since the lower R-height

$$r_-(x+2) = \frac{x+1}{2}$$

is already behind the current value $y = x+2$, the next R-point is the upper one,

$$r_+(x+2) = \frac{x+1}{2} + \frac{m}{2}.$$

Because $x \le m-7$, this lies strictly between $x+2$ and m. So

$$(x+2, x+2) \xrightarrow{G^{\frac{m-x-3}{2}}} (x+2, r_+(x+2)) \xrightarrow{R} (x+4, r_+(x+2)+2)\,.$$

The new lane $x+4$ is even, so there is no further R-branch. Its only A-point is at height $x+3$, already behind the current height $r_+(x+2)+2$. The B-branch occurs only at $y=0$ or at the isolated points $(0,1)$, $(m-3, m-2)$, $(m-2, m-1)$, none of which lies ahead on lane $x+4$. Hence every remaining step before the return to L_0 is generic, and therefore

$$T_0(x) = x+4.$$

The time is

$$\rho_0(x) = 1 + (x-1) + 1 + \frac{m-x-3}{2} + 1 + \frac{m-x-5}{2} = m-2.$$

Case x even, $4 \le x \le m-8$. The orbit begins

$$(x, 0) \xrightarrow{B} (x+1, 1),$$

and now the lane $x+1$ is odd. Its next R-height is the lower one

$$r_-(x+1) = \frac{x}{2},$$

so

$$(x+1, 1) \xrightarrow{G^{\frac{x}{2}-1}} \left(x+1, \frac{x}{2}\right) \xrightarrow{R} \left(x+3, \frac{x}{2}+2\right).$$

The new lane $x+3$ is odd. Its lower R-height is $x/2+1$, already behind the current height, while the A-height is $x+2$ and, because $x \le m-8$, this comes before the upper R-height. Thus

$$\left(x+3, \frac{x}{2}+2\right) \xrightarrow{G^{\frac{x}{2}}} (x+3, x+2) \xrightarrow{A} (x+4, x+4).$$

The lane $x+4$ is even, so there is no R-branch on it. Its only A-point is at height $x+3$, already behind the current height $x+4$, and the B-branch occurs only at $y=0$ or at the isolated points $(0,1)$, $(m-3, m-2)$, $(m-2, m-1)$, none of which lies ahead on lane $x+4$. So

$$T_0(x) = x+4,$$

and

$$\rho_0(x) = 1 + \left(\frac{x}{2} - 1\right) + 1 + \frac{x}{2} + 1 + (m-x-4) = m-2.$$

Case $x = m-6$. Here the orbit is

$$(m-6, 0) \xrightarrow{B} (m-5, 1) \xrightarrow{G^{\frac{m-8}{2}}} \left(m-5, \frac{m-6}{2}\right) \xrightarrow{R} \left(m-3, \frac{m}{2}-1\right) \xrightarrow{G^{\frac{m-6}{2}}} (m-3, m-4) \xrightarrow{A} (m-2, m-2)$$

$$\xrightarrow{Q}(1,1)\xrightarrow{S}(0,1)\xrightarrow{B}(1,2)\xrightarrow{G^{\frac{m-4}{2}}}\left(1,\frac{m}{2}\right)\xrightarrow{R}\left(3,\frac{m}{2}+2\right)\xrightarrow{G^{\frac{m-4}{2}}}(3,0).$$

Thus

$$T_0(m-6)=3,$$

and

$$\rho_0(m-6)=1+\frac{m-8}{2}+1+\frac{m-6}{2}+1+1+1+1+\frac{m-4}{2}+1+\frac{m-4}{2}=2m-4.$$

Case $x=m-5$**.** The orbit is

$$(m-5,0)\xrightarrow{B}(m-4,1)\xrightarrow{G^{m-6}}(m-4,m-5)\xrightarrow{A}(m-3,m-3)\xrightarrow{G}(m-3,m-2)\xrightarrow{B}(m-2,m-1)\xrightarrow{B}(m-1,0).$$

Therefore

$$T_0(m-5)=m-1,\qquad \rho_0(m-5)=1+(m-6)+1+1+1+1=m-1.$$

Case $x=m-4$**.** The orbit is

$$(m-4,0)\xrightarrow{B}(m-3,1)\xrightarrow{G^{\frac{m}{2}-3}}\left(m-3,\frac{m-4}{2}\right)\xrightarrow{R}\left(m-1,\frac{m}{2}\right)\xrightarrow{G^{\frac{m}{2}-2}}(m-1,m-2)\xrightarrow{A}(0,0).$$

So

$$T_0(m-4)=0,\qquad \rho_0(m-4)=1+\left(\frac{m}{2}-3\right)+1+\left(\frac{m}{2}-2\right)+1=m-2.$$

Case $x=m-3$**.** The orbit is

$$(m-3,0)\xrightarrow{B}(m-2,1)\xrightarrow{G^{m-4}}(m-2,m-3)\xrightarrow{A}(m-1,m-1)\xrightarrow{Q}(2,2)\xrightarrow{G^{m-2}}(2,0).$$

Hence

$$T_0(m-3)=2,\qquad \rho_0(m-3)=1+(m-4)+1+1+(m-2)=2m-3.$$

Case $x=m-2$**.** The orbit is

$$(m-2,0)\xrightarrow{R}(0,2)\xrightarrow{G^{m-3}}(0,m-1)\xrightarrow{R}(2,1)\xrightarrow{A}(3,3)\xrightarrow{G^{\frac{m-4}{2}}}\left(3,\frac{m}{2}+1\right)\xrightarrow{R}\left(5,\frac{m}{2}+3\right)\xrightarrow{G^{\frac{m-6}{2}}}(5,0).$$

Therefore

$$T_0(m-2)=5,$$

and

$$\rho_0(m-2)=1+(m-3)+1+1+\frac{m-4}{2}+1+\frac{m-6}{2}=2m-4.$$

Case $x=m-1$**.** The orbit is

$$(m-1,0)\xrightarrow{G^{\frac{m}{2}-1}}\left(m-1,\frac{m}{2}-1\right)\xrightarrow{R}\left(1,\frac{m}{2}+1\right)\xrightarrow{G^{\frac{m}{2}-1}}(1,0).$$

Hence

$$T_0(m-1)=1,\qquad \rho_0(m-1)=\left(\frac{m}{2}-1\right)+1+\left(\frac{m}{2}-1\right)=m-1.$$

These cases exhaust $x\in\mathbb{Z}_m$. □

Proposition 9 (splicing of arithmetic runs for color 0, Case II)**.** *Assume $m \equiv 4 \pmod 6$. Then T_0 is the successor permutation on a cyclic order obtained from four long arithmetic runs together with $O(1)$ bridge vertices. More precisely, when $m \equiv 10 \pmod{12}$ one has*

$$(0, m-2, 5, 9, \ldots, m-1, 1 \mid 4, 8, \ldots, m-6 \mid 3, 7, \ldots, m-3 \mid 2, 6, \ldots, m-4),$$

while when $m \equiv 4 \pmod{12}$ one has

$$(0, m-2, 5, 9, \ldots, m-3 \mid 2, 6, \ldots, m-6 \mid 3, 7, \ldots, m-1, 1 \mid 4, 8, \ldots, m-4).$$

Consequently T_0 is a single m-cycle.

Proof. Proposition 8 shows that, away from the boundary values

$$0, 1, 2, m-6, m-5, m-4, m-3, m-2, m-1,$$

the map acts by $x \mapsto x+4$. Thus the generic behavior is a disjoint union of long arithmetic runs modulo 4, and the exceptional values determine only how these runs are spliced together.

When $m \equiv 10 \pmod{12}$, the endpoint values give

$$0 \to m-2 \to 5, \qquad m-1 \to 1, \qquad m-6 \to 3, \qquad m-3 \to 2, \qquad m-4 \to 0,$$

while the generic rule $x \mapsto x+4$ propagates along the runs

$$5, 9, \ldots, m-1, \qquad 4, 8, \ldots, m-6, \qquad 3, 7, \ldots, m-3, \qquad 2, 6, \ldots, m-4.$$

This is exactly the first displayed cyclic concatenation.

When $m \equiv 4 \pmod{12}$, the endpoint values give

$$0 \to m-2 \to 5, \qquad m-3 \to 2, \qquad m-6 \to 3, \qquad m-1 \to 1, \qquad m-4 \to 0,$$

and the same generic rule $x \mapsto x+4$ propagates along the runs

$$5, 9, \ldots, m-3, \qquad 2, 6, \ldots, m-6, \qquad 3, 7, \ldots, m-1, \qquad 4, 8, \ldots, m-4.$$

So this time one obtains the second displayed cyclic concatenation.

In either congruence class Lemma 1 applies, and T_0 is a single m-cycle. □

Corollary 5. *If $m \equiv 4 \pmod 6$, then R_0 is a single m^2-cycle.*

Proof. Proposition 9 shows that T_0 is a single m-cycle. Also

$$\sum_{x \in \mathbb{Z}_m} \rho_0(x) = 1 + (m-6)(m-2) + 2(2m-4) + 2(m-1) + (2m-3) = m^2.$$

Lemma 7 finishes the proof. □

Corollary 6. *For every even $m \geq 6$, the return map R_0 is a single m^2-cycle on P_0.*

Proof. Combine Corollaries 4 and 5. □

D An explicit decomposition for $m = 4$

For $m = 4$, define a direction assignment by specifying at each vertex $(i, j, k) \in (\mathbb{Z}_4)^3$ a direction triple $(d_0, d_1, d_2) \in \{0, 1, 2\}^3$, where $0, 1, 2$ correspond respectively to the directions $+i, +j, +k$. The entry in the table for fixed i and coordinates (j, k) is the three-digit word $d_0 d_1 d_2$.

$i = 0$

j \ k	0	1	2	3
0	210	012	120	021
1	201	021	120	210
2	120	012	201	210
3	201	201	210	102

$i = 1$

j \ k	0	1	2	3
0	120	210	120	210
1	102	021	201	012
2	021	201	210	120
3	210	201	012	201

$i = 2$

j \ k	0	1	2	3
0	021	210	201	021
1	012	201	120	210
2	210	120	210	102
3	102	102	012	210

$i = 3$

j \ k	0	1	2	3
0	021	201	012	120
1	210	210	120	021
2	201	021	201	210
3	201	120	201	210

The direction table above is the finite witness. Because $D(4)$ has only 64 vertices, the table alone is sufficient for a complete finite verification: one can follow each color from any starting vertex and check that the orbit has length 64. For convenience, the accompanying file `m4_witness.json` records the direction table together with the three color orbits obtained by starting from $(0, 0, 0)$, and the script `verify_m4_witness.py` automates this finite audit:

1. it checks that every table entry is a permutation of $(0, 1, 2)$;
2. it follows each color from $(0, 0, 0)$ using the table and verifies that the orbit has length 64 before returning;
3. it confirms that the stored machine-readable cycle lists agree with those computed directly from the table.

This keeps the appendix readable while preserving a complete explicit witness for the boundary case $m = 4$.

E Verification artifacts

The verification artifacts focus on the even case $m \geq 6$, which involves the longest case analysis. The odd construction is given by a short closed form in the main text. For the boundary case $m = 4$, the same directory includes the finite witness `m4_witness.json` and its checker `verify_m4_witness.py`.

The main script `route_e_even.py` checks that, for every even m with $6 \leq m \leq 60$:

1. the resulting direction assignment is a valid coloring (each color is a permutation and the three outgoing arcs at every vertex are distinct);
2. the return maps on P_0 agree with the displayed piecewise formulas;
3. the induced return maps on P_0 are single m^2-cycles;

4. each color permutation is a single Hamilton cycle.

Two auxiliary scripts target the most verification-heavy steps in the appendices:

(a) `routee_return_formula_tables_check.py` checks the partition sets, low-layer words, and return-map formulas of Proposition 1;

(b) `routee_first_return_check.py` checks the appendix first-return formulas against the actual return-map dynamics on the stated transversals.

For larger even m, the main script can also validate the construction by checking only the closed-form first-return maps and return-time sums. These computations are not part of the proofs in the present paper; they serve as independent verification of the even-case construction and as regression checks for the displayed closed-form formulas.